

Numerical Analysis Assignment 1

nijatabbasov06

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1 1.1

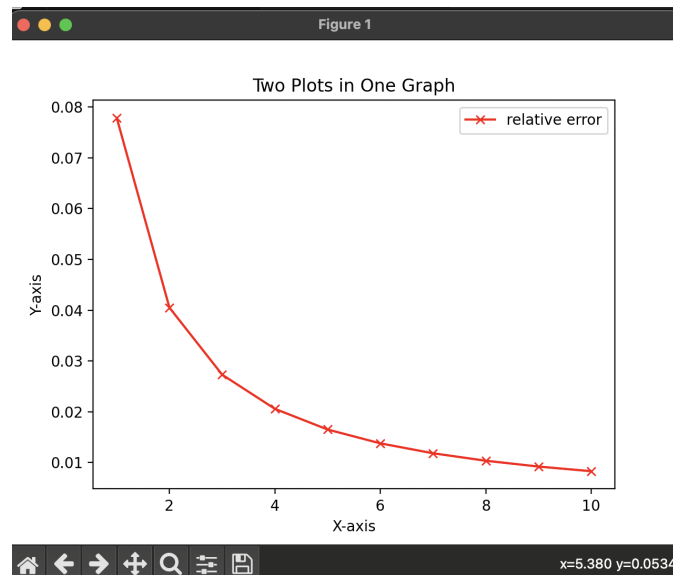


Figure 1: Relative error graph

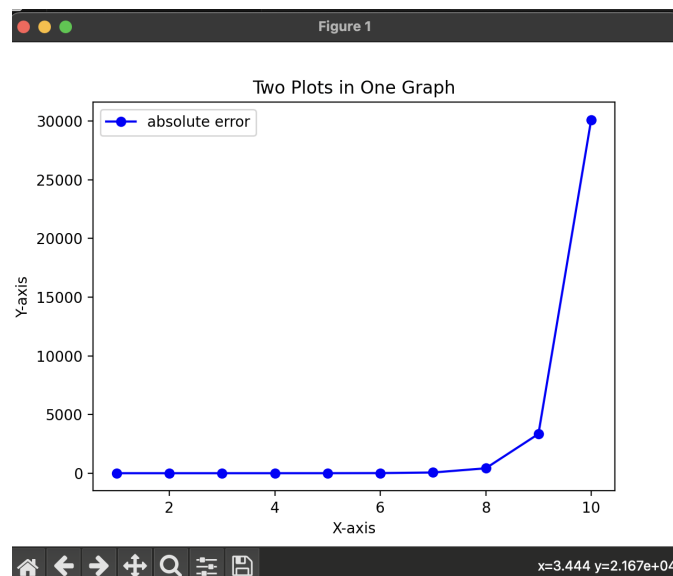
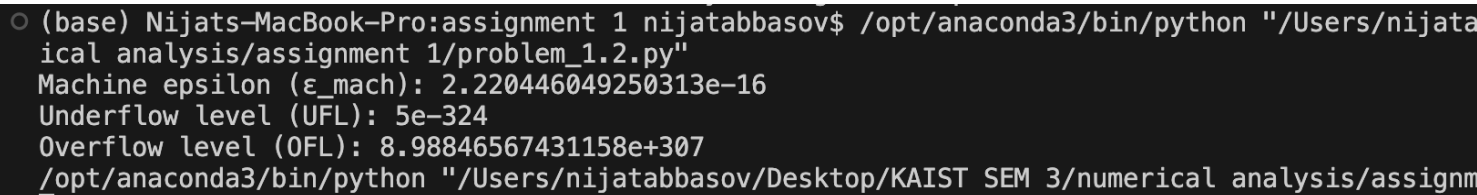


Figure 2: Absolute error graph

For the relative error graph, the error is shrinking as we are increasing n . To elaborate, sterling's approximation is better approximation for large n values.

Whereas for the absolute error graph, the error increases, as the absolute error is equal to relative error multiplied by $n!$. When n gets enormously large, the absolute error is getting bigger. But in relation to the value of n , as expected, relative error is getting smaller.

2 1.2



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○ (base) Nijats-MacBook-Pro:assignment 1 nijatabbasov$ /opt/anaconda3/bin/python "/Users/nijata
ical analysis/assignment 1/problem_1.2.py"
Machine epsilon ( $\epsilon_{\text{mach}}$ ): 2.220446049250313e-16
Underflow level (UFL): 5e-324
Overflow level (OFL): 8.98846567431158e+307
/opt/anaconda3/bin/python "/Users/nijatabbasov/Desktop/KAIST SEM 3/numerical analysis/assignm

```

Figure 3: Values of Machine Epsilon, Overflow and Underflow

To find the Machine Epsilon, we go inside a loop. First we add $s = 1$ to the value 1, if the result is not equal to 1, we divide s by 2 and add it to 1 again. If result is not equal to 1, we continue the process till the new generated s value is equal to 1. By this we find the Machine Epsilon.

To find the UFL, we divide the float 1.0 by 2 till the result is 0 and save the last result as UFL.

For OFL, we multiply float 1.0 by 2 till the result is NaN.

Moreover, we have 52 bits of mantissa (53 bits - 1 bit because of the leading 1 is not saved as a data), because I am using 64 bit computer and the 11 bit exponent. We can find these values from the underflow and overflow levels and the machine epsilon value. To elaborate, Machine epsilon $= 2^{-52}$. Also, by finding 2^{11} , we can find out the range of the values that can be saved. Which are UFL and OFL.

3 1.14

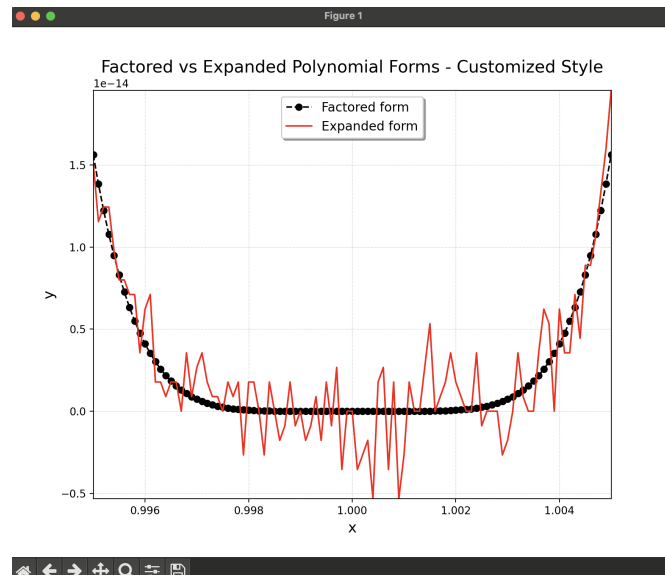


Figure 4: Factored and Expanded form

This behaviour arises because when we expand the form, we have operations such as $x^6 - 6x^5$, $15x^4 - 20x^3$, $15x^2 - 6x$. These operations give much more error than solely deducting 1 from x , because we are deducting two close values 3 times whereas for the $(x - 1)^6$ case we only deducted once. Moreover, while for $(x - 1)^6$ case we only took the power 6 times, we do this a lot of times for the expanded form which results in more error. Also, from the graph we can see that the error increases as x gets closer to 1. This is because the values that we are subtracting get closer to each other.

4 1.3.3

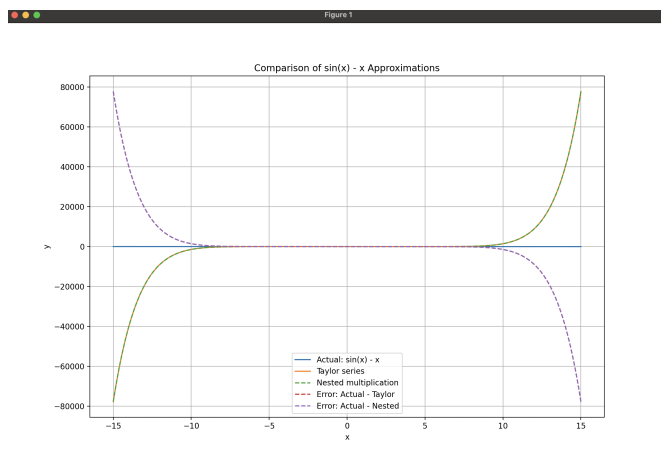


Figure 5: Graph for bigger x values

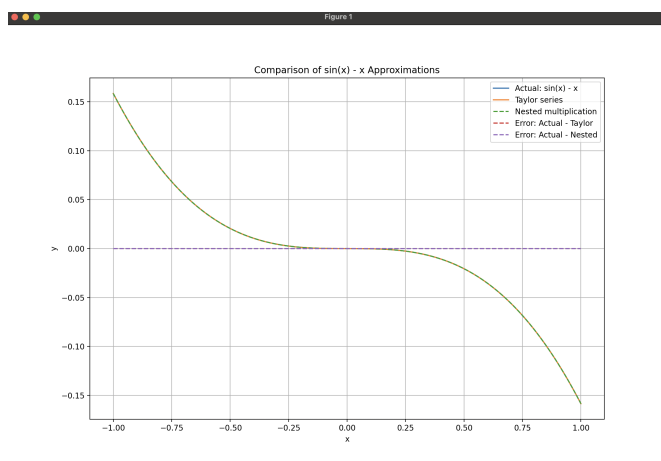


Figure 6: Graph for smaller x values

In the second graph, we can see that in the figures, as the value of x gets bigger, the difference between $\sin(x)$ and x gets bigger, as expected, because x is only an approximation for $\sin(x)$. Error between $\sin(x)$ and x is lower than 0.025 inside the interval $[-0.5; 0.5]$.

Now, let us compare the difference between nested multiplication, which is truncated and the actual series representation which is not truncated. In the second graph, we can see that, Taylor series and Nested

multiplication diverge as the value of x gets really big. The reason for this is that both of those graph represent truncated series. To elaborate, when the value of x gets big, the high order x terms are no longer negligible. So, truncating the series means, getting a lot of error. However, we can see that the error of Actual graph seems to be close to zero in relation to truncated series. So, we can say that the error due to truncation is much bigger than the error due to approximating $\sin(x)$ as x .